5G RAN

# Mobility Load Balancing in Idle Mode Feature Parameter Description

| | |
|---|---|
| **Issue** | Draft A |
| **Date** | 2025-12-31 |

HUAWEI TECHNOLOGIES CO., LTD.

## Trademarks and Permissions



## Notice



# Huawei Technologies Co., Ltd.

| | |
|---|---|
| Address: | Huawei Industrial Base<br>Bantian, Longgang<br>Shenzhen 518129<br>People's Republic of China |
| Website: | https://www.huawei.com |
| Email: | support@huawei.com |

# Security Declaration

## Vulnerability

Huawei's regulations on product vulnerability management are subject to the *Vul. Response Process*. For details about this process, visit the following web page:
https://www.huawei.com/en/psirt/vul-response-process
For vulnerability information, enterprise customers can visit the following web page:
https://securitybulletin.huawei.com/enterprise/en/security-advisory

# Contents



# 1 Change History

The following describes the specific changes in different issues.

## 5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 02 (2025-04-27).

- Technical changes

| Change Description | Parameter Change | RAT | Base Station Model |
|---|---|---|---|
| Added support for idle-mode MLB frequency optimization. For details, see:<br>• **Check of Whether the Cell and Frequency Requirements Are Met** in **6.1.2 MLB Procedure**<br>• **6.2.2 Impacts**<br>• **6.4.1.1 Data Preparation**<br>• **6.4.1.2 Using MML Commands** | Modified parameter: Added the **IDLE_MLB_FREQ_OPT_SW** option to the **NRCellMlb**.***MlbAlgoEnhSwitch*** parameter. | Low-frequency TDD | • 3900 and 5900 series base stations (macro base stations)<br>• DBS3900 LampSite and DBS5900 LampSite |

| Change Description | Parameter Change | RAT | Base Station Model |
|---|---|---|---|
| Added support for the configurable timer T320. For details, see:<br>• **RRC Connection Release** in **6.1.2 MLB Procedure**<br>• **6.3.2.3 Function Impacts**<br>• **6.4.1.1 Data Preparation**<br>• **6.4.1.2 Using MML Commands** | Added the **NRCellReselConfig**.***T320ForMlb*** parameter. | Low-frequency TDD | • 3900 and 5900 series base stations (macro base stations)<br>• DBS3900 LampSite and DBS5900 LampSite |
| Added support for high-frequency UE-number-based idle mode MLB by LampSite base stations. For details, see **7.3.3 Hardware**. | None | High-frequency TDD | DBS5900 LampSite |

- Editorial changes

  Revised descriptions of baseline priorities. For details, see **6.1.2 MLB Procedure**.

  Added descriptions of the mutually exclusive relationship between cell deployment in IAB mode and load balancing. For details, see **5.3.2.2 Mutually Exclusive Functions** and **6.3.2.2 Mutually Exclusive Functions**.

  Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

# 2 About This Document

## 2.1 General Statements

### Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

**NOTE**

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

### Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

## 2.2 Features in This Document

| Feature ID | Feature Name | Chapter/Section |
| --- | --- | --- |
| FOFD-051212 | Converged Load Balancing | **6 Low-Frequency Dedicated-Priority-based Idle Mode MLB** |

## 2.3 Differences

### Differences Between NSA and SA

| Function Name | Difference | Chapter/Section |
| --- | --- | --- |
| Low-frequency deprioritisation-based idle mode MLB | Supported only in SA networking | **5 Low-Frequency Deprioritisation-based Idle Mode MLB** |
| Low-frequency dedicated-priority-based idle mode MLB | Supported only in SA networking | **6 Low-Frequency Dedicated-Priority-based Idle Mode MLB** |
| High-frequency UE-number-based idle mode MLB (high-frequency TDD) | Supported only in SA networking | **7 High-Frequency UE-Number-based Idle Mode MLB (High-Frequency TDD)** |

### Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

| Function Name | Difference | Chapter/Section |
| --- | --- | --- |
| Low-frequency deprioritisation-based idle mode MLB | Supported only in low frequency bands | **5 Low-Frequency Deprioritisation-based Idle Mode MLB** |

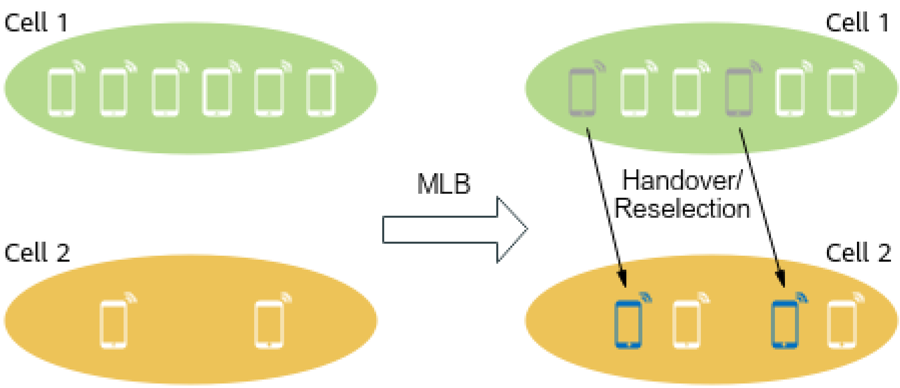
| Function Name | Difference | Chapter/Section |
|---|---|---|
| Low-frequency dedicated-priority-based idle mode MLB | Supported only in low frequency bands | **6 Low-Frequency Dedicated-Priority-based Idle Mode MLB** |
| High-frequency UE-number-based idle mode MLB (high-frequency TDD) | Supported only in high frequency bands: This function works only in SA-networking FWA scenarios. | **7 High-Frequency UE-Number-based Idle Mode MLB (High-Frequency TDD)** |

# 3 Overview

Among inter-frequency co-coverage cells in a 5G network, some cells may be heavily loaded while others are lightly loaded, which causes load imbalance. This imbalance will affect UE performance in high-load cells and waste resources in low-load ones.

To address this issue, mobility load balancing (MLB) is introduced. As shown in **Figure 3-1**, MLB balances cell loads by transferring UEs in connected or UEs in idle/inactive mode from high-load cells to low-load ones through handovers or reselections. This improves the resource utilization and capacity of cells involved in MLB.

**Figure 3-1** MLB




NOTE

Unless otherwise specified, a handover in this document refers to a serving cell change in SA networking and a PSCell change in NSA networking.

# 4 General Principles

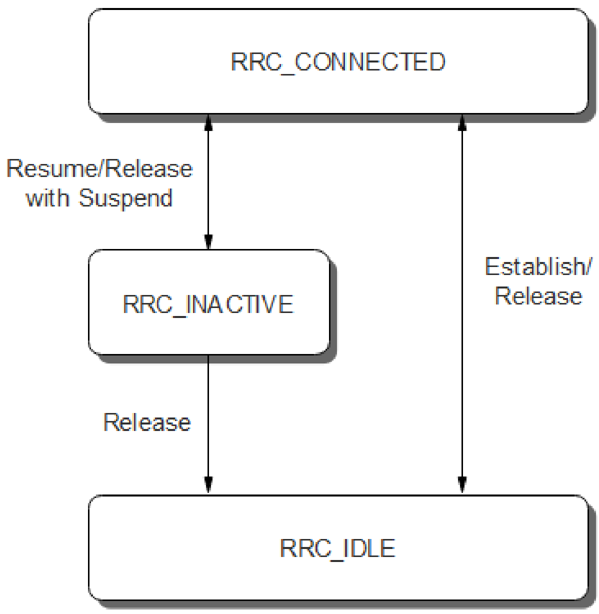
This chapter describes the general principles of MLB.

## 4.1 Basic Concepts

### UE States

In a 5G network, a UE is in one of the following radio resource control (RRC) states: RRC_CONNECTED, RRC_IDLE, and RRC_INACTIVE states. **Figure 4-1** illustrates the transitions between these RRC states.

**Figure 4-1** RRC states




### MLB Mode

MLB can be classified into the connected mode MLB and idle/inactive mode MLB functions to work on UEs in different states:

- Connected mode MLB: This function transfers UEs in connected mode from high-load cells to low-load ones through handovers to balance cell loads. For details about handover, see *SA Mobility Management in Connected Mode* and *NSA Mobility Management*.
- Idle/Inactive mode MLB: UEs in idle mode and inactive mode do not contribute to the cell load. This is because no RRC connection is set up between UEs in idle mode and the gNodeB and data processing for UEs in inactive mode is suspended. Idle/Inactive mode MLB transfers UEs in idle mode and inactive mode from high-load cells to low-load ones through reselections. In this way, UEs will not enter connected mode in the high-load cells, and thus will not further increase load of the high-load cells. For details about cell reselection, see *SA Mobility Management in Idle Mode*.

  **NOTE**

  Both idle mode MLB and inactive mode MLB are implemented through cell reselection, the principles of which are the same. This document uses idle mode MLB as an example to offer corresponding descriptions. For details about whether UEs in inactive mode support certain idle mode MLB functions, see **4.3.3 MLB Overview**.

### MLB Direction

MLB involves the following directions:

- Uplink MLB: performed based on the uplink load of a cell only. Uplink MLB only transfers uplink UEs from high-load cells to low-load cells.
- Downlink MLB: performed based on the downlink load of a cell only. Downlink MLB only transfers downlink UEs from high-load cells to low-load cells.
- Uplink and downlink MLB: performed based on both the uplink and downlink loads of a cell. Uplink and downlink MLB transfer both uplink and downlink UEs from high-load cells to low-load cells.

**NOTE**

For the definitions of uplink and downlink loads, see **4.2 Load Evaluation Dimensions**. For the definitions of uplink and downlink UEs, see **4.2.1 Number of UEs**.

## 4.2 Load Evaluation Dimensions

The load of a cell is evaluated from the following dimensions:

- **4.2.1 Number of UEs**
- **4.2.2 Air Interface Load of a Cell**
- **4.2.3 Number of 5QI-specific UEs**
- **4.2.4 5QI-specific UE-Number-based Air Interface Load**
- **4.2.5 Equivalent CCE Usage**

During the MLB procedure, the gNodeB evaluates the load of a cell based on one or more load evaluation dimensions.

## 4.2.1 Number of UEs

The number of UEs in RRC_CONNECTED mode in a cell is used to measure the cell load:

- Number of UEs in the downlink in a cell, including:
  - Average number of UEs in RRC_CONNECTED mode in a cell (including non-CA UEs and CA UEs that treat the local cell as their PCell), which can be observed using the **N.User.RRCConn.Avg** counter
  - Average number of downlink CA UEs that treat the local cell as their SCell, which can be observed using the **N.User.CA.SCell.DL.Avg** counter
- Number of UEs in the uplink in a cell, including:
  - Average number of UEs in RRC_CONNECTED mode in a cell (including non-CA UEs and CA UEs that treat the local cell as their PCell), which can be observed using the **N.User.RRCConn.Avg** counter
  - Average number of uplink CA UEs that treat the local cell as their SCell, which can be observed using the **N.User.CA.SCell.UL.Avg** counter
  - Average number of UEs that treat the local cell as their SUL cell, which can be observed using the **N.User.SUL.UL.Avg** counter

**NOTE**

As UEs in idle mode and inactive mode have no impact on cell load, the number of UEs in connected mode, instead of those in idle or inactive mode, in a cell is still used to measure the cell load in idle mode MLB.

Unless otherwise stated, the number of UEs defined in this section applies throughput this document.

## 4.2.2 Air Interface Load of a Cell

The air interface load of a cell reflects the spectral efficiency of the cell. The air interface load of a cell is calculated as follows:

- Downlink air interface load of a cell = Number of downlink UEs in the cell/ Downlink air interface capability of the cell
- Uplink air interface load of a cell = Number of uplink UEs in the cell/Uplink air interface capability of the cell

The air interface capability of a cell is related to the load evaluation policy used for MLB:

- Static evaluation (The **NRCellMlb**.***LoadEvaluationPolicy*** parameter is set to **STATIC_EVAL**.)
  - Downlink air interface capability of a cell = (Cell bandwidth in the unit of MHz x Proportion of downlink slots in the cell x Downlink cell capability scale factor)/1000. (Downlink cell capability scale factor is specified by the **NRCellMlb**.***CellDlCapbScaleFactor*** parameter.)
  - Uplink air interface capability of a cell = (Cell bandwidth in the unit of MHz x Proportion of uplink slots in the cell x Uplink cell capability scale factor)/1000. (Uplink cell capability scale factor is specified by the **NRCellMlb**.***CellUlCapbScaleFactor*** parameter.)
- Dynamic evaluation (The **NRCellMlb**.***LoadEvaluationPolicy*** parameter is set to **DYNAMIC_EVAL**.)

- Downlink air interface capability of a cell = (Downlink cell traffic volume x Number of available PRBs for the downlink data channel/Number of PRBs actually occupied by the downlink data channel)/100000
- Uplink air interface capability of a cell = (Uplink cell traffic volume x Number of available PRBs for the uplink data channel/Number of PRBs actually occupied by the uplink data channel)/100000

If the number of UEs in a cell is small (less than 20) or the PRB usage of the cell is low (less than 20% in the uplink or less than 10% in the downlink) for a long time, static evaluation is recommended. In other scenarios, dynamic evaluation is recommended. When the dynamic evaluation policy is used and the number of UEs or the PRB usage is small in a short period, the gNodeB automatically switches to the static evaluation policy to calculate the air interface capability of a cell.

**NOTE**

- Proportion of downlink slots in a cell: NR TDD: Proportion of downlink slots in a cell = (Number of downlink-only slots + Number of self-contained slots)/(Number of downlink-only slots + Number of self-contained slots + Number of uplink-only slots). For example, if the slot configuration of a cell is 4:1, the proportion of downlink slots in the cell is 0.8 (4/5).
- Proportion of uplink slots in a cell: NR TDD: Proportion of uplink slots in a cell = Number of uplink-only slots/(Number of downlink-only slots + Number of self-contained slots + Number of uplink-only slots). For example, if the slot configuration of a cell is 4:1, the proportion of uplink slots in the cell is 0.2 (1/5).

## 4.2.3 Number of 5QI-specific UEs

The number of 5QI-specific UEs refers to the number of UEs running services with a 5QI specified by the **NRCellMlb**.***MlbUe5qi*** parameter in a cell.

- Number of 5QI-specific downlink UEs: Number of UEs running downlink services with a 5QI specified by the **NRCellMlb**.***MlbUe5qi*** parameter in a cell
- Number of 5QI-specific uplink UEs: Number of UEs running uplink services with a 5QI specified by the **NRCellMlb**.***MlbUe5qi*** parameter in a cell

**NOTE**

If inter-gNodeB MLB is triggered, the **gNB5qiConfig**.***CounterObjSubscriptionInd*** parameter must be set to **YES** on the gNodeB serving the inter-gNodeB neighboring cells for the 5QI specified by the **NRCellMlb**.***MlbUe5qi*** parameter of the serving cell.

## 4.2.4 5QI-specific UE-Number-based Air Interface Load

- 5QI-specific UE-number-based downlink air interface load = Number of UEs running downlink services with a 5QI specified by the **NRCellMlb**.***MlbUe5qi*** parameter in a cell/Downlink air interface capability of a cell

  Downlink air interface capability of a cell = Downlink cell traffic volume x Number of available PRBs on the downlink data channel/Number of PRBs occupied by the downlink data channel
- 5QI-specific UE-number-based uplink air interface load = Number of UEs running uplink services with a 5QI specified by the **NRCellMlb**.***MlbUe5qi*** parameter in a cell/Uplink air interface capability of a cell

Uplink air interface capability of a cell = Uplink cell traffic volume x Number of available PRBs on the uplink data channel/Number of PRBs occupied by the uplink data channel

### 4.2.5 Equivalent CCE Usage

The equivalent CCE usage is calculated as follows:

- Downlink equivalent CCE usage = **N.CCE.DLDCI.Used.Equivalent**/(**N.CCE.Avail.Equivalent - N.CCE.ULDCI.Avail.Equivalent**)

  That is: Downlink equivalent CCE usage = Number of PDCCH CCEs whose power is equivalent to the reference power and that are used by the downlink DCI/(Number of available PDCCH CCEs with power equivalent to the reference power - Number of PDCCH CCEs whose power is equivalent to the reference power and that are available for the uplink DCI)
- Uplink equivalent CCE usage = **N.CCE.ULDCI.Used.Equivalent**/**N.CCE.ULDCI.Avail.Equivalent**

  That is: Uplink equivalent CCE usage = Number of PDCCH CCEs whose power is equivalent to the reference power and that are used by the uplink DCI/Number of PDCCH CCEs whose power is equivalent to the reference power and that are available for the uplink DCI

## 4.3 MLB Procedure

### 4.3.1 Connected Mode MLB

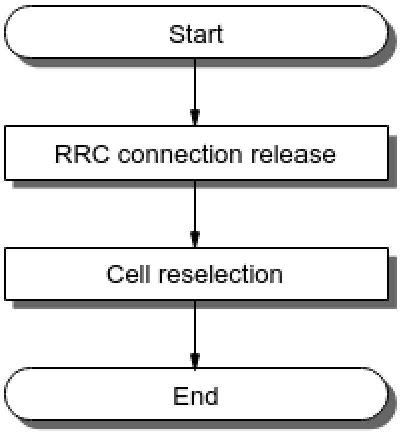
For details about the procedure for connected mode MLB, see *Mobility Load Balancing in Connected Mode*.

### 4.3.2 Idle Mode MLB

**Figure 4-2** shows the basic procedure for idle mode MLB, including the following steps:

- RRC connection release: After the RRC connection is released, the UE enters idle mode.
- Cell reselection: The UE reselects a cell for access based on the policy delivered by the gNodeB. For details about cell reselection, see, *SA Mobility Management in Idle Mode*.

**Figure 4-2** Basic procedure for idle mode MLB




Idle mode MLB in low-frequency scenarios is controlled by the **INTER_FREQ_IDLE_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter, with the load evaluation policy specified by the different settings of the **NRCellMlb**.***IdleMlbPolicy*** parameter as follows:

- If the **NRCellMlb**.***IdleMlbPolicy*** parameter is set to **DECREASE_PRIORITY**, low-frequency deprioritisation-based idle mode MLB is enabled. For details, see **5.1 Principles**.
- If the **NRCellMlb**.***IdleMlbPolicy*** parameter is set to **DEDICATED_PRIORITY**, low-frequency dedicated-priority-based idle mode MLB is enabled, and the corresponding sub-functions can be enabled by different settings of the **NRCellMlb**.***MlbTriggerMode*** parameter as follows:
  - UE-number-based idle mode MLB can be enabled by setting it to **CELL_USER_NUM**.
  - Spectral-efficiency-based idle mode MLB can be enabled by setting it to **SPEC_EFF_BASED_USER_NUM**.
  - Radio-resource-based idle mode MLB can be enabled by setting it to **RADIO_RESOURCE**. Radio resources involve the number of UEs in connected mode, the equivalent CCE usage, and other aspects.

  Before RRC connections are released for low-frequency dedicated-priority-based idle mode MLB, other steps such as the candidate cell selection, inter-cell load information exchange, as well as frequency selection and sequencing are performed. For details, see **6.1 Principles**.

In high-frequency TDD scenarios, idle mode MLB is controlled by the **INTER_FREQ_IDLE_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter. This function balances the number of UEs in connected mode among common cells (cells with the **NRDUCell**.***SupplementaryCellInd*** parameter set to **NORMAL_CELL**) in a sector. For details, see **7.1 Principles**.

## 4.3.3 MLB Overview

MLB mainly provides the following functions:

- Connected mode MLB: For details, see *Mobility Load Balancing in Connected Mode*.

- Idle mode MLB: For details, see **Table 4-1**.

**Table 4-1** Idle mode MLB overview

| MLB Function | Application Scenario | Supported in Inactive Mode? |
| --- | --- | --- |
| **5 Low-Frequency Deprioritisation-based Idle Mode MLB** | Low-frequency TDD | Yes |
| **6 Low-Frequency Dedicated-Priority-based Idle Mode MLB** | Low-frequency TDD | Yes |
| **7 High-Frequency UE-Number-based Idle Mode MLB (High-Frequency TDD)** | High-frequency TDD | No |

# 5 Low-Frequency Deprioritisation-based Idle Mode MLB

## 5.1 Principles

In low-frequency deprioritisation-based idle mode MLB, the number of UEs in connected mode in a cell is used to indicate the cell load. When the gNodeB determines that the loads are unbalanced between cells, it will transfer UEs in idle mode between cells through reselections to achieve load balancing. When this function takes effect, the gNodeB sets the cell-reselection priority for the operating frequency of the local cell to the lowest (valid for 30 minutes) and delivers the priority to UEs through the *deprioritisationReq* IE in the RRCRelease message. This avoids UEs from re-accessing the local cell within a short period of time.

Low-frequency deprioritisation-based idle mode MLB requires the following configurations:

- **NRCellMlb**.***MlbTriggerMode*** set to **CELL_USER_NUM**
- **NRCellMlb**.***IdleMlbPolicy*** set to **DECREASE_PRIORITY**
- **INTER_FREQ_IDLE_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter selected

It is recommended that this function be enabled when all of the following conditions are met:

- The same proportions of small-packet and large-packet services are independently running in inter-frequency co-coverage cells.
- Most UEs are motionless or move slowly.

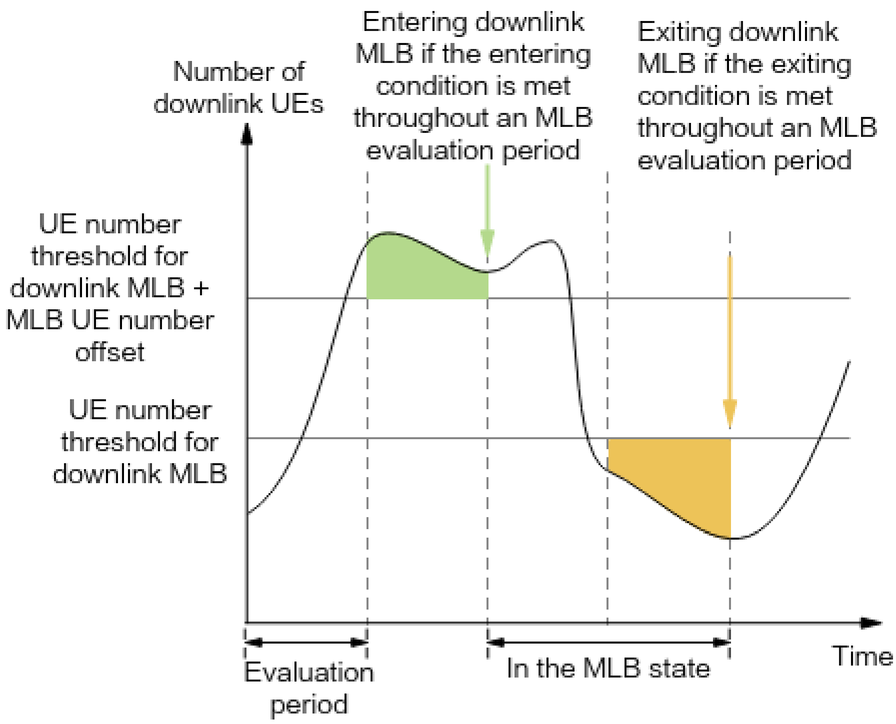
### 5.1.1 Entering and Exiting MLB

After low-frequency deprioritisation-based idle mode MLB is enabled, the gNodeB determines whether a cell enters or exits the MLB state in the next MLB evaluation period based on the cell load within the current period. Based on the MLB direction, MLB is divided into the following:

- Downlink MLB
  - Entering condition: The number of UEs in the downlink is greater than or equal to the sum of the inter-frequency MLB downlink UE number threshold and MLB downlink UE number offset throughout an MLB evaluation period.
  - Exiting condition: The number of UEs in the downlink is less than the inter-frequency MLB downlink UE number threshold throughout an MLB evaluation period.
- Uplink MLB
  - Entering condition: The number of UEs in the uplink is greater than or equal to the sum of the inter-frequency MLB uplink UE number threshold and MLB uplink UE number offset throughout an MLB evaluation period.
  - Exiting condition: The number of UEs in the uplink is less than the inter-frequency MLB uplink UE number threshold throughout an MLB evaluation period.
- Uplink and downlink MLB
  - Entering condition: The entering condition for the uplink or downlink MLB state is met.
  - Exiting condition: The exiting conditions for both the uplink and downlink MLB states are met.

If the gNodeB determines that a cell meets the conditions for entering or exiting the MLB state in the current MLB evaluation period, the cell enters or exits the MLB state in the next MLB evaluation period. **Figure 5-1** uses downlink MLB as an example and shows the conditions for entering and exiting downlink MLB.

**Figure 5-1** Conditions for entering and exiting downlink MLB



NOTE

For details about the definition of the MLB direction, see **MLB Direction** in **4.1 Basic Concepts**. For details about the definitions of the numbers of UEs in the downlink and uplink, see **4.2.1 Number of UEs**.

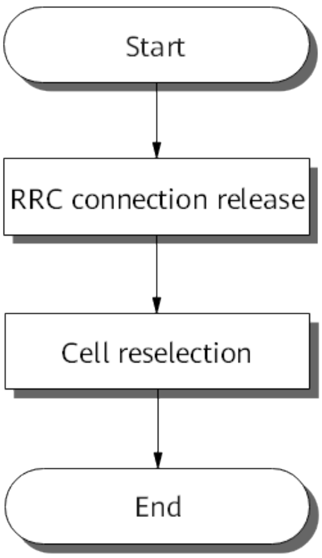
The following describes the parameters involved in the conditions for entering and exiting MLB.

- The MLB evaluation period is specified by the **NRCellMlb**.***MlbJudgePeriod*** parameter.
- The MLB direction is configured using the **NRCellMlb**.***MlbDirection*** parameter.
  - **DOWNLINK**: indicates that downlink MLB is enabled.
  - **UPLINK**: indicates that uplink MLB is enabled.
  - **DUAL**: indicates that uplink and downlink MLB is enabled.
- The inter-frequency MLB downlink UE number threshold is specified by the **NRCellMlb**.***InterFreqMlbUeNumThld*** parameter.
- The MLB downlink UE number offset is specified by the **NRCellMlb**.***MlbUeNumOffset*** parameter.
- The inter-frequency MLB uplink UE number threshold is specified by the **NRCellMlb**.***InterFreqMlbUlUeNumThld*** parameter.
- The MLB uplink UE number offset is specified by the **NRCellMlb**.***MlbUlUeNumOffset*** parameter.

## 5.1.2 MLB Procedure

After a cell enters the MLB state, MLB is performed as shown in **Figure 5-2**.

**Figure 5-2** MLB procedure




### RRC Connection Release

The gNodeB sets the cell-reselection priority for the operating frequency of the local cell to the lowest within a valid period of 30 minutes and delivers the priority to UEs through the *deprioritisationReq* IE in the RRCRelease message. The maximum number of UEs that can be selected is specified by the **NRCellMlb**.***IdleMlbUeNumThld*** parameter. The *deprioritisationReq* IE contains the following information:

- Deprioritisation type: frequency
- Deprioritisation period: fixed at 30 minutes

After the RRC connection of a UE is released, the UE enters the idle mode.

NOTE

Low-frequency deprioritisation-based idle mode MLB does not take effect for UEs with RFSPs delivered from the core network or UEs for which RRC connection releases are not triggered.

In low-frequency scenarios, if the **CA_UE_FREQ_STEER_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter is selected in the serving cell of a CA-capable UE, the gNodeB determines whether to filter out the frequencies of neighboring cells based on the UE's CA capability and the frequency and networking continuity of the serving cell on which the UE camps, when delivering frequencies and frequency priorities. The frequencies that are not filtered out are delivered to the UE. The UE can reselect to the corresponding frequencies based on the dedicated reselection priorities. For details about frequency filtering scenarios and recommended scenarios for frequency steering for CA UEs, see *SA Mobility Management in Connected Mode*.

### Cell Reselection

The UE reselects to a cell according to the policy delivered by the gNodeB. For details about cell reselection, see, *SA Mobility Management in Idle Mode*.

# 5.2 Network Analysis

## 5.2.1 Benefits

Low-frequency deprioritisation-based idle mode MLB increases the resource usage and capacity of cells involved in MLB. The higher the serving cell's load and the greater the inter-cell load difference, the more the gains.

The following must also be considered when using low-frequency deprioritisation-based idle mode MLB:

- When UEs in the inter-frequency co-coverage cells are high-mobility UEs, this function brings neither positive gains nor negative impacts. Therefore, it is recommended that this function be disabled.
- If the numbers of UEs in inter-frequency co-coverage cells are not independent, for example, when CA between two cells is performed for a large number of UEs, this function brings neither positive gains nor negative impacts. Therefore, it is recommended that this function be disabled.
- The gains of this function are reduced when cells on multiple frequencies differ greatly in their coverage capabilities, UEs differ greatly in their band support capabilities, or there are restrictions on the serving PLMNs of UEs due to UE subscription.

## 5.2.2 Impacts

The impacts between this function and other functions are described in **5.3.2.3 Function Impacts**.

After low-frequency deprioritisation-based idle mode MLB is enabled, the gNodeB includes the *deprioritisationReq* IE in the RRCRelease message to reduce the priority of the operating frequency to the lowest. If there is no neighboring NR frequency, the UE will reselect to an LTE cell.

# 5.3 Requirements

## 5.3.1 Licenses

None

## 5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

### 5.3.2.1 Prerequisite Functions

None

#### 5.3.2.2 Mutually Exclusive Functions

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | 5QI-specific UE-number-based connected mode MLB | **INTER_FREQ_CONNECTED_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter selected and the **NRCellMlb**.***MlbTriggerMode*** parameter set to **SPECIFIED_5QI_USER_NUM**. | *Mobility Load Balancing in Connected Mode* | If the **INTER_FREQ_IDLE_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter is selected, the **NRCellMlb**.***MlbTriggerMode*** parameter cannot be set to **SPECIFIED_5QI_USER_NUM**. |
| Low-frequency TDD | Cell deployment in IAB mode | **NRDUCell**.***DeployPosition*** set to **DEPLOY_IAB** | None | Idle mode MLB (controlled by the **INTER_FREQ_IDLE_MLB_SW** option of the **RCellAlgoSwitch**.***MlbAlgoSwitch*** parameter) cannot be enabled for cells deployed in IAB mode. |

#### 5.3.2.3 Function Impacts

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RFSP-based camping and handover policies | **gNBRfspConfig**.***gNBFreqPriorityGroupId*** | *Flexible User Steering* | Low-frequency deprioritisation-based idle mode MLB does not take effect for UEs configured with RFSP-specific dedicated cell-reselection priorities. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD<br>High-frequency TDD | Differentiated carrier selection for FWA users | **NRCellQciBearer**.***gNBFreqPriorityGroupId*** set to a value ranging from **0** to **254** | *FWA* | In SA networking, low-frequency deprioritisation-based idle mode MLB does not take effect for UEs for which differentiated carrier selection for FWA UEs has taken effect. |

## 5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

### Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

### Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

### RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

## 5.3.4 Networking

There must be at least two frequencies available on the live network.

## 5.3.5 Others

None

# 5.4 Operation and Maintenance

## 5.4.1 Data Configuration

### 5.4.1.1 Data Preparation

**Table 5-1** and **Table 5-2** describe the parameters used for function activation and optimization, respectively. This section does not describe parameters related to cell establishment.

**Table 5-1** Parameters used for activation

| Parameter Name | Parameter ID | Option | Setting Notes |
|---|---|---|---|
| MLB Algorithm Switch | **NRCellAlgoSwitch**.***MlbAlgoSwitch*** | INTER_FREQ_IDLE_MLB_SW | Select this option. |
| Idle MLB Policy | **NRCellMlb**.***IdleMlbPolicy*** | None | Set this parameter to **DECREASE_PRIORITY**. |
| MLB Trigger Mode | **NRCellMlb**.***MlbTriggerMode*** | None | Set this parameter to **CELL_USER_NUM**. |

**Table 5-2** Parameters used for optimization

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| MLB Judge Period | **NRCellMlb**.***MlbJudgePeriod*** | Use the recommended value. |
| MLB Direction | **NRCellMlb**.***MlbDirection*** | Set this parameter based on the network plan. |
| Inter-frequency MLB UE Number Threshold | **NRCellMlb**.***InterFreqMlbUeNumThld*** | Set this parameter based on the network plan. |
| Inter-frequency MLB UL UE Number Thld | **NRCellMlb**.***InterFreqMlbUlUeNumThld*** | Set this parameter based on the network plan. |
| MLB UE Number Offset | **NRCellMlb**.***MlbUeNumOffset*** | Set this parameter based on the network plan. |
| MLB UL UE Number Offset | **NRCellMlb**.***MlbUlUeNumOffset*** | Set this parameter based on the network plan. |

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| Idle MLB UE Num Threshold | **NRCellMlb.*IdleMlbUeNumThld*** | Set this parameter based on the network plan.<br>The value of this threshold is evenly allocated to the user management modules of a cell. If the allocated value is a decimal, the value will be rounded up. In this case, the total number of UEs in the cell for which MLB can be performed will exceed the value of this threshold. |

### 5.4.1.2 Using MML Commands

Before using MML commands, refer to **5.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

## Activation Command Examples

```
//Turning on the MLB algorithm switch for a cell
MOD NRCELLALGOSWITCH: NrCellId=0, MlbAlgoSwitch=INTER_FREQ_IDLE_MLB_SW-1;
//Setting the idle-mode MLB trigger mode and setting the idle mode MLB policy to the deprioritisation-based policy
MOD NRCELLMLB: NrCellId=0, MlbTriggerMode=CELL_USER_NUM, IdleMlbPolicy=DECREASE_PRIORITY;
```

## Optimization Command Examples

```
//Setting the MLB evaluation period for the cell
MOD NRCELLMLB: NrCellId=0, MlbJudgePeriod=10;
//Setting the MLB direction for the cell
MOD NRCELLMLB: NrCellId=0, MlbDirection=DOWNLINK;
//Setting the InterFreqMlbUeNumThld and MlbUeNumOffset parameters
MOD NRCELLMLB: NrCellId=0, InterFreqMlbUeNumThld=200, MlbUeNumOffset=50;
//Setting the InterFreqMlbUlUeNumThld and MlbUlUeNumOffset parameters
MOD NRCELLMLB: NrCellId=0, InterFreqMlbUlUeNumThld=100, MlbUlUeNumOffset=20;
//Setting the IdleMlbUeNumThld parameter
MOD NRCELLMLB: NrCellId=0, MlbTriggerMode=SPEC_EFF_BASED_USER_NUM, IdleMlbUeNumThld=100;
```

## Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the MLB algorithm switch for the cell
MOD NRCELLALGOSWITCH: NrCellId=0, MlbAlgoSwitch=INTER_FREQ_IDLE_MLB_SW-0;
```

### 5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

**NOTE**

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

## 5.4.2 Activation Verification

After low-frequency deprioritisation-based idle mode MLB is enabled, if it is observed that the *deprioritisationReq* IE in the RRCRelease message contains the information about the deprioritisation type and duration and the values of the following counters become relatively balanced between cells, this function has taken effect.

**Table 5-3** MLB-related counters for a cell

| Counter ID | Counter Name |
|---|---|
| 1911816771 | **N.User.RRCConn.Avg** |
| 1911816773 | **N.User.CA.PCell.DL.Avg** |
| 1911816775 | **N.User.CA.PCell.UL.Avg** |

On the MAE-Access, start a signaling trace management task with the parameters for **MLB Monitoring** under **Cell Performance Monitoring** specified to monitor MLB. If either bit 0 or bit 5 of the binary number converted from the decimal value of **MLB Status** is **1**, MLB has taken effect.

## 5.4.3 Network Monitoring

Observe the value of the **N.User.RRCConn.Avg** counter to check whether the distribution of UEs in connected mode in the cells on different frequencies meets the expectation of MLB.

**NOTE**

As low-frequency deprioritisation-based idle mode MLB does not take effect for UEs for which RRC connection releases are not triggered, the **N.User.RRCConn.Avg** counter value still differs among intra-sector cells, although the numbers of UEs among cells are balanced.

On the MAE-Access, start a signaling trace management task with the parameters for **MLB Monitoring** under **Cell Performance Monitoring** specified to monitor MLB. The number of times idle mode MLB takes effect on the gNodeB can be monitored by **Idle Mlb Trigger Num**.

# 6 Low-Frequency Dedicated-Priority-based Idle Mode MLB

## 6.1 Principles

In low-frequency dedicated-priority-based idle mode MLB, indicators such as the number of UEs in connected mode in a cell and the equivalent CCE usage are used to indicate the cell load. When the base station determines that the loads are unbalanced between cells, it will transfer UEs in idle mode to cells on low-load frequencies to achieve a load balance.

Assume that the **NRCellMlb**.***IdleMlbPolicy*** parameter is set to **DEDICATED_PRIORITY** and the **INTER_FREQ_IDLE_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter is selected. The corresponding sub-functions of low-frequency dedicated-priority-based idle mode MLB can be enabled through different settings of the **NRCellMlb**.***MlbTriggerMode*** parameter as follows:

- UE-number-based idle mode MLB can be enabled by setting it to **CELL_USER_NUM**.
- Spectral-efficiency-based idle mode MLB can be enabled by setting it to **SPEC_EFF_BASED_USER_NUM**.
- Radio-resource-based idle mode MLB can be enabled by setting it to **RADIO_RESOURCE**. Radio resources involve the number of UEs in connected mode, the equivalent CCE usage, and other aspects. The **RADIO_RESOURCE** value can be used only in low-frequency TDD.

### 6.1.1 Entering and Exiting MLB

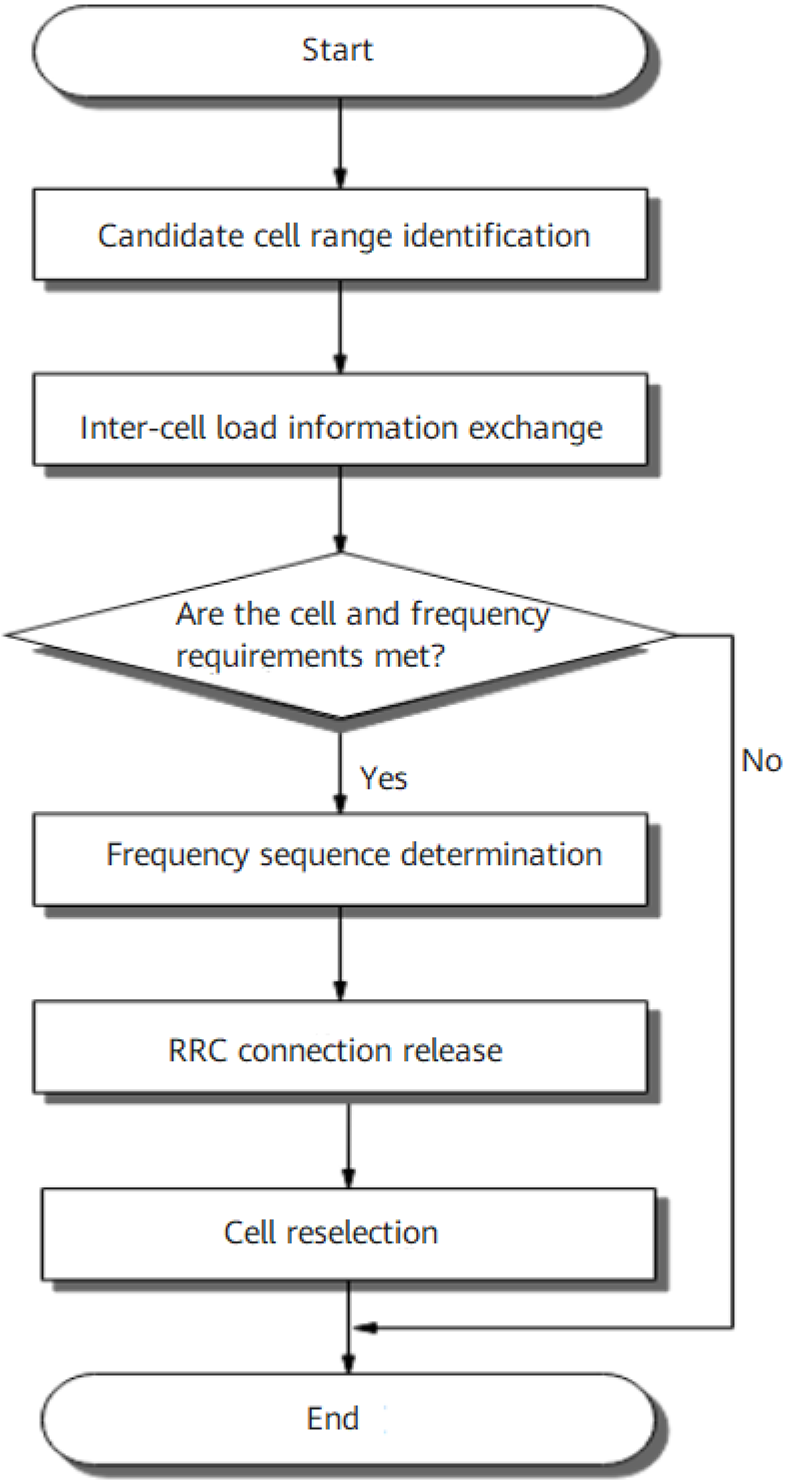
- The conditions for entering and exiting UE-number-based idle mode MLB are the same as those for UE-number-based connected mode MLB. For details, see descriptions related to UE-number-based connected mode MLB in *Mobility Load Balancing in Connected Mode*.
- The conditions for entering and exiting spectral-efficiency-based idle mode MLB are the same as those for spectral-efficiency-based connected mode MLB. For details, see descriptions related to spectral-efficiency-based connected mode MLB in *Mobility Load Balancing in Connected Mode*.

- The conditions for entering and exiting radio-resource-based idle mode MLB are the same as those for radio-resource-based connected mode MLB. For details, see descriptions related to radio-resource-based connected mode MLB in *Mobility Load Balancing in Connected Mode*.

### 6.1.2 MLB Procedure

After a cell enters the MLB state, MLB is performed as shown in **Figure 6-1**.

**Figure 6-1** MLB procedure




## Candidate Cell Range Identification

- Candidate cell range identification for UE-number-based idle mode MLB is the same as that for UE-number-based connected mode MLB. For details, see descriptions related to UE-number-based connected mode MLB in *Mobility Load Balancing in Connected Mode*.

- Candidate cell selection for spectral-efficiency-based idle mode MLB is the same as that for spectral-efficiency-based connected mode MLB. For details, see descriptions related to spectral-efficiency-based connected mode MLB in *Mobility Load Balancing in Connected Mode*.
- Candidate cell selection for radio-resource-based idle mode MLB is the same as that for radio-resource-based connected mode MLB. For details, see descriptions related to radio-resource-based connected mode MLB in *Mobility Load Balancing in Connected Mode*.

## Inter-Cell Load Information Exchange

- Inter-cell load information exchange for UE-number-based idle mode MLB is the same as that for UE-number-based connected mode MLB. For details, see descriptions related to UE-number-based connected mode MLB in *Mobility Load Balancing in Connected Mode*.
- Inter-cell load information exchange for spectral-efficiency-based idle mode MLB is the same as that for spectral-efficiency-based connected mode MLB. For details, see descriptions related to spectral-efficiency-based connected mode MLB in *Mobility Load Balancing in Connected Mode*.
- Inter-cell load information exchange for radio-resource-based idle mode MLB is the same as that for radio-resource-based connected mode MLB. For details, see descriptions related to radio-resource-based connected mode MLB in *Mobility Load Balancing in Connected Mode*.

## Check of Whether the Cell and Frequency Requirements Are Met

**A candidate cell meeting all of the following conditions is considered as a qualified low-load cell:**

- The load difference between the local cell and the candidate cell is greater than the load difference threshold. The threshold and determination method of load difference for idle mode MLB are as follows:
  - The load difference determination method for UE-number-based idle mode MLB is the same as that for UE-number-based connected mode MLB. For details, see descriptions related to UE-number-based connected mode MLB in *Mobility Load Balancing in Connected Mode*.
  - The load difference determination method for spectral-efficiency-based idle mode MLB is the same as that for spectral-efficiency-based connected mode MLB. For details, see descriptions related to spectral-efficiency-based connected mode MLB in *Mobility Load Balancing in Connected Mode*.
  - The load difference determination method for radio-resource-based idle mode MLB is basically the same as that for radio-resource-based connected mode MLB. For details, see descriptions related to radio-resource-based connected mode MLB in *Mobility Load Balancing in Connected Mode*. Their differences are as follows:
    - In radio-resource-based connected mode MLB, the base station looks for a neighboring cell that meets the load difference condition in the candidate cell list based on the number of UEs in connected mode first. If no qualified cell exists, then the base station looks again based on the equivalent CCE usage instead.

    - In radio-resource-based idle mode MLB, the base station considers a neighboring cell in the candidate cell list as a qualified candidate cell if it meets the load difference condition based on either the number of UEs in connected mode or equivalent CCE usage.

- The candidate cell is not in the high-load state. (The cell load must be less than **NRCellMlb**.***MlbHoAdmissionThld***.)

Dedicated-priority-based idle mode MLB will not be triggered if no candidate cell is lightly loaded, and the procedure ends.

**The load on a neighboring frequency is evaluated as follows:**

- If the **NRCellRelation**.***MlbHoFlag*** parameter is set to **TRUE** for any cell on a neighboring frequency, the following evaluation applies:
  – In case that all the cells for which the **NRCellRelation**.***MlbHoFlag*** parameter is set to **TRUE** are lightly loaded, the neighboring frequency is considered as a low-load frequency.
  – Otherwise, the neighboring frequency is considered as a high-load frequency.
- If the **NRCellRelation**.***MlbHoFlag*** parameter is set to **FALSE** for all cells on a neighboring frequency, the following evaluation applies:
  – In case that idle-mode MLB frequency optimization is enabled, the frequency is considered as a high-load frequency.
  – In case that idle-mode MLB frequency optimization is disabled, the frequency is considered as a low-load frequency if any cell on the frequency is lightly loaded; otherwise, the frequency is considered as a high-load frequency.

  Idle-mode MLB frequency optimization is controlled by the **IDLE_MLB_FREQ_OPT_SW** option of the **NRCellMlb**.***MlbAlgoEnhSwitch*** parameter.

Dedicated-priority-based idle mode MLB will not be triggered if no low-load frequency exists, and the procedure ends.

## Frequency Sequencing

The current frequency and all neighboring frequencies are sequenced in descending order of priority as follows:

1. Low-load neighboring NR frequencies (If multiple such frequencies exist, they are sequenced based on their baseline priorities.)
2. The current NR frequency
3. Neighboring NR frequencies whose load information is not obtained and high-load NR frequencies (If multiple such frequencies exist, they are sequenced based on their baseline priorities.)
4. Neighboring LTE frequencies (If multiple such frequencies exist, they are sequenced based on their baseline priorities.)

Baseline priorities include dedicated and common cell-reselection priorities. For details, see the "Cell Reselection Priority" section in *SA Mobility Management in Idle Mode*.

NOTE

If there is no NR low-load frequency among the preceding frequencies, dedicated-priority-based idle mode MLB will not be triggered and the procedure ends.

### RRC Connection Release

The gNodeB sends the sequenced frequencies and frequency priorities to the to-be-released UEs through the *CellReselectionPriorities* IE in the RRCRelease message. At the same time, the gNodeB starts the T320 timer (configured by the **NRCellReselConfig**.***T320ForMlb*** parameter). When the timer expires, the to-be-released UEs resume the system-provided common reselection priorities. The maximum number of selected UEs is specified by the **NRCellMlb**.***IdleMlbUeNumThld*** parameter.

After the RRC connection of a UE is released, the UE enters the idle mode.

NOTE

Low-frequency dedicated-priority-based idle mode MLB does not take effect for UEs for which the RRC connection releases are not triggered.

In low-frequency scenarios, if the **CA_UE_FREQ_STEER_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter is selected in the serving cell of a CA-capable UE, the gNodeB determines whether to filter out the frequencies of neighboring cells based on the UE's CA capability and the frequency and networking continuity of the serving cell on which the UE camps, when delivering frequencies and frequency priorities. The frequencies that are not filtered out are delivered to the UE. The UE can reselect to the corresponding frequencies based on the dedicated reselection priorities. For details about frequency filtering scenarios and recommended scenarios for frequency steering for CA UEs, see *SA Mobility Management in Connected Mode*.

### Cell Reselection

The UE reselects to a cell according to the policy delivered by the gNodeB. For details about cell reselection, see *SA Mobility Management in Idle Mode*.

# 6.2 Network Analysis

## 6.2.1 Benefits

Low-frequency dedicated-priority-based idle mode MLB increases the resource usage and capacity of cells involved in MLB. The higher the cell load and the greater the inter-cell load difference, the more the gains.

The gains of this function are reduced when cells on multiple frequencies differ greatly in their coverage capabilities, UEs differ greatly in their band support, or there are restrictions on the serving PLMNs of UEs due to UE subscription.

## 6.2.2 Impacts

The impacts between this function and other functions are described in **6.3.2.3 Function Impacts**.

After idle-mode MLB frequency optimization is enabled, the number of target frequencies available for idle mode MLB may decrease, which in turn may cause low-frequency dedicated-priority-based idle mode MLB to yield fewer gains.

# 6.3 Requirements

## 6.3.1 Licenses

### License Requirements for UE-Number-based Idle Mode MLB

None

### License Requirements for Spectral-Efficiency-based Idle Mode MLB

| Feature ID | Feature Name | Model | Sales Unit |
|---|---|---|---|
| FOFD-051212 | Converged Load Balancing | NR0S000CLB00 | per Cell |

**NOTE**

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

### License Requirements for Radio-Resource-based Idle Mode MLB

| Feature ID | Feature Name | Model | Sales Unit |
|---|---|---|---|
| FOFD-051212 | Converged Load Balancing | NR0S000CLB00 | Per Cell |

**NOTE**

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

## 6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

#### 6.3.2.1 Prerequisite Functions

None

#### 6.3.2.2 Mutually Exclusive Functions

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | 5QI-specific UE-number-based connected mode MLB | **INTER_FREQ_CONNECTED_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter selected, and the **NRCellMlb**.***MlbTriggerMode*** parameter set to **SPECIFIED_5QI_USER_NUM**. | *Mobility Load Balancing in Connected Mode* | If the **INTER_FREQ_IDLE_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter is selected, the **NRCellMlb**.***MlbTriggerMode*** parameter cannot be set to **SPECIFIED_5QI_USER_NUM**. |
| Low-frequency TDD | Cell deployment in IAB mode | **NRDUCell**.***DeployPosition*** set to **DEPLOY_IAB** | None | Idle mode MLB (controlled by the **INTER_FREQ_IDLE_MLB_SW** option of the **RCellAlgoSwitch**.***MlbAlgoSwitch*** parameter) cannot be enabled for cells deployed in IAB mode. |

#### 6.3.2.3 Function Impacts

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Inter-frequency MLB switch for idle mode | **INTER_FREQ_IDLE_MLB_SW** option of the **gNBRfspConfig**.***MlbAlgoSwitch*** parameter | *Mobility Load Balancing in Idle Mode* | If inter-frequency idle mode MLB is enabled, UEs with RFSPs configured can be transferred to inter-frequency cells through idle mode MLB. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD<br>High-frequency TDD | Differentiated carrier selection for FWA users | **NRCellQciBearer**.***gNBFreqPriorityGroupId*** set to a value ranging from **0** to **254** | *FWA* | In SA networking, low-frequency dedicated-priority-based idle mode MLB does not take effect for UEs with differentiated carrier selection for FWA UEs in effect. |
| Low-frequency TDD | Normal-UE frequency steering | **NORMAL_UE_FREQ_STEER_SW** option of the **NRCellMntPara**.***MntIntrfCtrlAlgoSw*** parameter | None | When normal-UE frequency steering is enabled together with low-frequency dedicated-priority-based idle mode MLB, only the former takes effect. |
| Low-frequency TDD | Handover optimization for normal UEs | **NORMAL_UE_HO_OPT_SW** option of the **NRCellMntPara**.***MntIntrfCtrlAlgoSw*** parameter | None | When handover optimization for normal UEs is enabled together with low-frequency dedicated-priority-based idle mode MLB, only the former takes effect. |
| Low-frequency TDD | High-speed Railway Superior Experience | **NRDUCell**.***HighSpeedFlag*** | *High Speed Mobility* | If High-speed Railway Superior Experience is enabled together with low-frequency dedicated-priority-based idle mode MLB and the **NRCellFreqRelation**.***IdleMlbUeReleaseRatio*** parameter is set to **0**, only the former takes effect. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Reselection policy for 3CC and massive CA UEs in idle mode | **MASS_CA_IDLE_RESEL_POL_SW** option of the **NRCellOpQciBearer**.***ServiceDiffAlgoSwitch*** parameter | *CA Experience Improvement* | If both dedicated-priority-based idle mode MLB (**NRCellMlb**.***IdleMlbPolicy*** set to **DEDICATED_PRIORITY**, with the **INTER_FREQ_IDLE_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter being selected) and the reselection policy for 3CC and massive CA UEs in idle mode are enabled, the T320 timer takes the value of **NRCellReselConfig**.***T320ForOther*** for UEs in the 3CC aggregation state and massive CA UEs. For details about **NRCellReselConfig**.***T320ForOther***, see *SA Mobility Management in Idle Mode*. |
| Low-frequency TDD | T320 Timer For Other | **NRCellReselConfig**.***T320ForOther*** | *SA Mobility Management in Idle Mode* | |

## 6.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

### Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

### Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

### RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

## 6.3.4 Networking

There must be at least two frequencies available on the live network.

## 6.3.5 Others

None

# 6.4 Operation and Maintenance

## 6.4.1 Data Configuration

### 6.4.1.1 Data Preparation

**Table 6-1** describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

**Table 6-1** Parameters used for activation

| Parameter Name | Parameter ID | Option | Setting Notes |
|---|---|---|---|
| MLB Algorithm Switch | **NRCellAlgoSwitch.** ***MlbAlgoSwitch*** | INTER_FREQ_IDLE_MLB_SW | Select this option. |
| Idle MLB Policy | **NRCellMlb.*IdleMlbPolicy*** | None | Set this parameter to **DEDICATED_PRIORITY**. |

**Table 6-2** describes the parameters related to UE-number-based idle mode MLB.

**Table 6-2** Parameters related to UE-number-based idle mode MLB

| Parameter Name | Parameter ID | Option | Setting Notes |
| --- | --- | --- | --- |
| MLB Trigger Mode | **NRCellMlb**.***MlbTriggerMode*** | None | Set this parameter to **CELL_USER_NUM**. |
| MLB Judge Period | **NRCellMlb**.***MlbJudgePeriod*** | None | Use the recommended value. |
| MLB Direction | **NRCellMlb**.***MlbDirection*** | None | Set this parameter based on the network plan. |
| Inter-frequency MLB UE Number Threshold[a] | **NRCellMlb**.***InterFreqMlbUeNumThld*** | None | Set this parameter based on the network plan. |
| MLB UE Number Offset[a] | **NRCellMlb**.***MlbUeNumOffset*** | None | Set this parameter based on the network plan. |
| Inter-frequency MLB UL UE Number Thld[a] | **NRCellMlb**.***InterFreqMlbUlUeNumThld*** | None | Set this parameter based on the network plan. |
| MLB UL UE Number Offset[a] | **NRCellMlb**.***MlbUlUeNumOffset*** | None | Set this parameter based on the network plan. |
| MLB-Based Handover Flag | **NRCellRelation**.***MlbHoFlag*** | None | Set this parameter based on the network plan. |
| MLB Enhancement Algorithm Switch | **NRCellMlb**.***MlbAlgoEnhSwitch*** | IDLE_MLB_FREQ_OPT_SW | If the **NRCellRelation**.***MlbHoFlag*** parameter is set to **FALSE** for all cells on a neighboring frequency, you are advised to select the **IDLE_MLB_FREQ_OPT_SW** option of this parameter to avoid cell reselections to the neighboring frequency through idle mode MLB. |

| Parameter Name | Parameter ID | Option | Setting Notes |
|---|---|---|---|
| No Handover Flag | **NRCellRelation**.***NoHoFlag*** | None | Set this parameter based on the network plan. |
| MLB HO Admission Thld | **NRCellMlb**.***MlbHoAdmissionThld*** | None | Set this parameter based on the network plan. |
| UE Number Difference Threshold | **NRCellMlb**.***UeNumDiffThld*** | None | Set this parameter based on the network plan. |
| UL UE Number Difference Threshold | **NRCellMlb**.***UlUeNumDiffThld*** | None | Set this parameter based on the network plan. |
| Connected MLB UE Num Upper Limit | **NRCellMlb**.***ConnMlbUeNumThld*** | None | Set this parameter based on the network plan.<br>If the **NRCellMlb**.***MlbJudgePeriod*** parameter is set to a value less than **10** (in the unit of second), it is recommended that the **NRCellMlb**.***ConnMlbUeNumThld*** parameter value be less than or equal to the **NRCellMlb**.***MlbJudgePeriod*** parameter value. |
| T320 Timer For Mlb | **NRCellReselConfig**.***T320ForMlb*** | None | Set this parameter based on the network plan. |

| Parameter Name | Parameter ID | Option | Setting Notes |
|---|---|---|---|
| a: The configuration reference is as follows:<br>• **NRCellMlb**.***InterFreqMlbUeNumThld***: Recommended value = Reference value of the UE number threshold (downlink) x 52/60<br>• **NRCellMlb**.***MlbUeNumOffset***: Recommended value = Reference value of the UE number threshold (downlink) x 8/60<br>• **NRCellMlb**.***InterFreqMlbUlUeNumThld***: Recommended value = Reference value of the UE number threshold (uplink) x 52/60<br>• **NRCellMlb**.***MlbUlUeNumOffset***: Recommended value = Reference value of the UE number threshold (uplink) x 8/60<br>If the busy-hour PRB usage of every cell that is planned to be involved in MLB is greater than or equal to 60%:<br>• Reference value of the UE number threshold (downlink) = **N.User.RRCConn.Avg** (when the **Downlink Resource Block Utilizing Rate (DU)** is equal to 60%)<br>• Reference value of the UE number threshold (uplink) = **N.User.RRCConn.Avg** (when the **Uplink Resource Block Utilizing Rate (DU)** is equal to 60%)<br>If the busy-hour PRB usage of the local cell that is planned to be involved in MLB is always less than 60%, and there are other cells whose PRB usage is greater than or equal to 60%, then the cell with the highest PRB usage among the other cells is selected as the reference cell.<br>• Reference value of the UE number threshold (uplink/downlink) = Number of UEs in the reference cell when the PRB usage (uplink/downlink) is 60% x Air interface capability of the local cell (uplink/downlink)/Air interface capability of the reference cell (uplink/downlink). Specifically:<br>  – Uplink air interface capability = **N.ThpVol.UL.Cell**/(**Uplink Resource Block Utilizing Rate (DU)**/100000)<br>  – Downlink air interface capability = **N.ThpVol.DL.Cell**/(**Downlink Resource Block Utilizing Rate (DU)**/100000)<br>If the busy-hour PRB usage of every cell is less than 60%, the uplink and downlink UE number thresholds are set to their default values for the cell with the highest uplink and downlink bandwidths. The reference values of the UE number thresholds for cells with other bandwidths are calculated as follows:<br>• Reference value of the UE number threshold for a cell with another bandwidth (uplink) = (Default value of the **NRCellMlb**.***InterFreqMlbUlUeNumThld*** parameter + Default value of the **NRCellMlb**.***MlbUlUeNumOffset*** parameter) x (Uplink bandwidth of the cell/Maximum uplink cell bandwidth)<br>• Reference value of the UE number threshold for a cell with another bandwidth (downlink) = (Default value of the **NRCellMlb**.***InterFreqMlbUeNumThld*** parameter + Default value of the **NRCellMlb**.***MlbUeNumOffset*** parameter) x (Downlink bandwidth of the cell/Maximum downlink cell bandwidth) | | | |

**Table 6-3** describes the parameters related to spectral-efficiency-based idle mode MLB.

**Table 6-3** Parameters related to spectral-efficiency-based idle mode MLB

| Parameter Name | Parameter ID | Option | Setting Notes |
| --- | --- | --- | --- |
| MLB Trigger Mode | **NRCellMlb**.***MlbTriggerMode*** | None | Set this parameter to **SPEC_EFF_BASED_USER_NUM**. |
| MLB Judge Period | **NRCellMlb**.***MlbJudgePeriod*** | None | Use the recommended value. |
| MLB Direction | **NRCellMlb**.***MlbDirection*** | None | Set this parameter based on the network plan. |
| Inter-frequency MLB UE Number Threshold[a] | **NRCellMlb**.***InterFreqMlbUeNumThld*** | None | Set this parameter based on the network plan. |
| MLB UE Number Offset[a] | **NRCellMlb**.***MlbUeNumOffset*** | None | Set this parameter based on the network plan. |
| Inter-frequency MLB UL UE Number Thld[a] | **NRCellMlb**.***InterFreqMlbUlUeNumThld*** | None | Set this parameter based on the network plan. |
| MLB UL UE Number Offset[a] | **NRCellMlb**.***MlbUlUeNumOffset*** | None | Set this parameter based on the network plan. |
| MLB-Based Handover Flag | **NRCellRelation**.***MlbHoFlag*** | None | Set this parameter based on the network plan. |
| MLB Enhancement Algorithm Switch | **NRCellMlb**.***MlbAlgoEnhSwitch*** | IDLE_MLB_FREQ_OPT_SW | If the **NRCellRelation**.***MlbHoFlag*** parameter is set to **FALSE** for all cells on a neighboring frequency, you are advised to select the **IDLE_MLB_FREQ_OPT_SW** option of this parameter to avoid cell reselections to the neighboring frequency. |

| Parameter Name | Parameter ID | Option | Setting Notes |
|---|---|---|---|
| No Handover Flag | **NRCellRelation**.***NoHoFlag*** | None | Set this parameter based on the network plan. |
| MLB HO Admission Thld | **NRCellMlb**.***MlbHoAdmissionThld*** | None | Set this parameter based on the network plan. |
| UE Number Difference Threshold | **NRCellMlb**.***UeNumDiffThld*** | None | Set this parameter based on the network plan. |
| UL UE Number Difference Threshold | **NRCellMlb**.***UlUeNumDiffThld*** | None | Set this parameter based on the network plan. |
| Load Evaluation Policy | **NRCellMlb**.***LoadEvaluationPolicy*** | None | The value **DYNAMIC_EVAL** is recommended. |
| Cell DL Capability Scale Factor | **NRCellMlb**.***CellDlCapbScaleFactor*** | None | Set this parameter based on the network plan. |
| Cell UL Capability Scale Factor | **NRCellMlb**.***CellUlCapbScaleFactor*** | None | Set this parameter based on the network plan. |
| MLB UE Selection Policy | **NRCellMlb**.***MlbUeSelectionPolicy*** | SA_USER | Set this parameter based on the network plan. |
| MLB UE Selection Policy | **NRCellMlb**.***MlbUeSelectionPolicy*** | NSA_USER | Set this option based on the network plan. |

| Parameter Name | Parameter ID | Option | Setting Notes |
|---|---|---|---|
| Connected MLB UE Num Upper Limit | **NRCellMlb**.***ConnMlbUeNumThld*** | None | Set this parameter based on the network plan.<br>If the **NRCellMlb**.***MlbJudgePeriod*** parameter is set to a value less than **10** (in the unit of second), it is recommended that the **NRCellMlb**.***ConnMlbUeNumThld*** parameter value be less than or equal to the **NRCellMlb**.***MlbJudgePeriod*** parameter value. |
| T320 Timer For Mlb | **NRCellReselConfig**.***T320ForMlb*** | None | Set this parameter based on the network plan. |

| Parameter Name | Parameter ID | Option | Setting Notes |
| --- | --- | --- | --- |

a: The configuration reference is as follows:

- **NRCellMlb**.***InterFreqMlbUeNumThld***: Recommended value = Reference value of the UE number threshold (downlink) x 52/60
- **NRCellMlb**.***MlbUeNumOffset***: Recommended value = Reference value of the UE number threshold (downlink) x 8/60
- **NRCellMlb**.***InterFreqMlbUlUeNumThld***: Recommended value = Reference value of the UE number threshold (uplink) x 52/60
- **NRCellMlb**.***MlbUlUeNumOffset***: Recommended value = Reference value of the UE number threshold (uplink) x 8/60

If the busy-hour PRB usage of every cell that is planned to be involved in MLB is greater than or equal to 60%:

- Reference value of the UE number threshold (downlink) = **N.User.RRCConn.Avg** (when the **Downlink Resource Block Utilizing Rate (DU)** is equal to 60%)
- Reference value of the UE number threshold (uplink) = **N.User.RRCConn.Avg** (when the **Uplink Resource Block Utilizing Rate (DU)** is equal to 60%)

If the busy-hour PRB usage of the local cell that is planned to be involved in MLB is always less than 60%, and there are other cells whose PRB usage is greater than or equal to 60%, then the cell with the highest PRB usage among the other cells is selected as the reference cell.

- Reference value of the UE number threshold (uplink/downlink) = Number of UEs in the reference cell when the PRB usage (uplink/downlink) is 60% x Air interface capability of the local cell (uplink/downlink)/Air interface capability of the reference cell (uplink/downlink). Specifically:
  - Uplink air interface capability = **N.ThpVol.UL.Cell**/(**Uplink Resource Block Utilizing Rate (DU)**/100000)
  - Downlink air interface capability = **N.ThpVol.DL.Cell**/(**Downlink Resource Block Utilizing Rate (DU)**/100000)

If the busy-hour PRB usage of every cell is less than 60%, the uplink and downlink UE number thresholds are set to their default values for the cell with the highest uplink and downlink bandwidths. The reference values of the UE number thresholds for cells with other bandwidths are calculated as follows:

- Reference value of the UE number threshold for another cell (uplink) = (Default value of the **NRCellMlb**.***InterFreqMlbUlUeNumThld*** parameter + Default value of the **NRCellMlb**.***MlbUlUeNumOffset*** parameter) x (Uplink bandwidth of the cell/Maximum uplink cell bandwidth)
- Reference value of the UE number threshold for another cell (downlink) = (Default value of the **NRCellMlb**.***InterFreqMlbUeNumThld*** parameter + Default value of the **NRCellMlb**.***MlbUeNumOffset*** parameter) x (Downlink bandwidth of the cell/Maximum downlink cell bandwidth)

**Table 6-4** describes the parameters related to radio-resource-based idle mode MLB.

**Table 6-4** Parameters related to radio-resource-based idle mode MLB

| Parameter Name | Parameter ID | Option | Setting Notes |
|---|---|---|---|
| MLB Trigger Mode | **NRCellMlb**.***MlbTriggerMode*** | None | Set this parameter to **RADIO_RESOURCE**. |
| MLB Judge Period | **NRCellMlb**.***MlbJudgePeriod*** | None | Use the recommended value. |
| MLB Direction | **NRCellMlb**.***MlbDirection*** | None | Set this parameter based on the network plan. |
| Inter-frequency MLB UE Number Threshold[a] | **NRCellMlb**.***InterFreqMlbUeNumThld*** | None | Set this parameter based on the network plan. |
| MLB UE Number Offset[a] | **NRCellMlb**.***MlbUeNumOffset*** | None | Set this parameter based on the network plan. |
| Inter-frequency MLB UL UE Number Thld[a] | **NRCellMlb**.***InterFreqMlbUlUeNumThld*** | None | Set this parameter based on the network plan. |
| MLB UL UE Number Offset[a] | **NRCellMlb**.***MlbUlUeNumOffset*** | None | Set this parameter based on the network plan. |
| MLB-Based Handover Flag | **NRCellRelation**.***MlbHoFlag*** | None | Set this parameter based on the network plan. |
| MLB Enhancement Algorithm Switch | **NRCellMlb**.***MlbAlgoEnhSwitch*** | IDLE_MLB_FREQ_OPT_SW | If the **NRCellRelation**.***MlbHoFlag*** parameter is set to **FALSE** for all cells on a neighboring frequency, you are advised to select the **IDLE_MLB_FREQ_OPT_SW** option of this parameter to avoid cell reselections to the neighboring frequency. |

| Parameter Name | Parameter ID | Option | Setting Notes |
|---|---|---|---|
| No Handover Flag | **NRCellRelation**.***NoHoFlag*** | None | Set this parameter based on the network plan. |
| MLB HO Admission Thld | **NRCellMlb**.***MlbHoAdmissionThld*** | None | Set this parameter based on the network plan. |
| UE Number Difference Threshold | **NRCellMlb**.***UeNumDiffThld*** | None | Set this parameter based on the network plan. |
| UL UE Number Difference Threshold | **NRCellMlb**.***UlUeNumDiffThld*** | None | Set this parameter based on the network plan. |
| Load Evaluation Policy | **NRCellMlb**.***LoadEvaluationPolicy*** | None | The value **DYNAMIC_EVAL** is recommended. |
| Cell DL Capability Scale Factor | **NRCellMlb**.***CellDlCapbScaleFactor*** | None | Set this parameter based on the network plan. |
| Cell UL Capability Scale Factor | **NRCellMlb**.***CellUlCapbScaleFactor*** | None | Set this parameter based on the network plan. |
| Connected MLB UE Num Upper Limit | **NRCellMlb**.***ConnMlbUeNumThld*** | None | Set this parameter based on the network plan.<br>If the **NRCellMlb**.***MlbJudgePeriod*** parameter is set to a value less than **10** (in the unit of second), it is recommended that the **NRCellMlb**.***ConnMlbUeNumThld*** parameter value be less than or equal to the **NRCellMlb**.***MlbJudgePeriod*** parameter value. |
| MLB DL CCE Usage Threshold | **NRCellMlb**.***MlbDlCceUsageThld*** | None | Set this parameter based on the network plan. |

| Parameter Name | Parameter ID | Option | Setting Notes |
|---|---|---|---|
| MLB UL CCE Usage Threshold | **NRCellMlb**.***MlbUlCceUsageThld*** | None | Set this parameter based on the network plan. |
| T320 Timer For Mlb | **NRCellReselConfig**.***T320ForMlb*** | None | Set this parameter based on the network plan. |

| Parameter Name | Parameter ID | Option | Setting Notes |
|---|---|---|---|
| a: The configuration reference is as follows:<br>• **NRCellMlb**.***InterFreqMlbUeNumThld***: Recommended value = Reference value of the UE number threshold (downlink) x 52/60<br>• **NRCellMlb**.***MlbUeNumOffset***: Recommended value = Reference value of the UE number threshold (downlink) x 8/60<br>• **NRCellMlb**.***InterFreqMlbUlUeNumThld***: Recommended value = Reference value of the UE number threshold (uplink) x 52/60<br>• **NRCellMlb**.***MlbUlUeNumOffset***: Recommended value = Reference value of the UE number threshold (uplink) x 8/60<br>If the busy-hour PRB usage of every cell that is planned to be involved in MLB is greater than or equal to 60%:<br>• Reference value of the UE number threshold (downlink) = **N.User.RRCConn.Avg** (when the **Downlink Resource Block Utilizing Rate (DU)** is equal to 60%)<br>• Reference value of the UE number threshold (uplink) = **N.User.RRCConn.Avg** (when the **Uplink Resource Block Utilizing Rate (DU)** is equal to 60%)<br>If the busy-hour PRB usage of the local cell that is planned to be involved in MLB is always less than 60%, and there are other cells whose PRB usage is greater than or equal to 60%, then the cell with the highest PRB usage among the other cells is selected as the reference cell.<br>• Reference value of the UE number threshold (uplink/downlink) = Number of UEs in the reference cell when the PRB usage (uplink/downlink) is 60% x Air interface capability of the local cell (uplink/downlink)/Air interface capability of the reference cell (uplink/downlink). Specifically:<br>  – Uplink air interface capability = **N.ThpVol.UL.Cell**/(**Uplink Resource Block Utilizing Rate (DU)**/100000)<br>  – Downlink air interface capability = **N.ThpVol.DL.Cell**/(**Downlink Resource Block Utilizing Rate (DU)**/100000)<br>If the busy-hour PRB usage of every cell is less than 60%, the uplink and downlink UE number thresholds are set to their default values for the cell with the highest uplink and downlink bandwidths. The reference values of the UE number thresholds for cells with other bandwidths are calculated as follows:<br>• Reference value of the UE number threshold for another cell (uplink) = (Default value of the **NRCellMlb**.***InterFreqMlbUlUeNumThld*** parameter + Default value of the **NRCellMlb**.***MlbUlUeNumOffset*** parameter) x (Uplink bandwidth of the cell/Maximum uplink cell bandwidth)<br>• Reference value of the UE number threshold for another cell (downlink) = (Default value of the **NRCellMlb**.***InterFreqMlbUeNumThld*** parameter + Default value of the **NRCellMlb**.***MlbUeNumOffset*** parameter) x (Downlink bandwidth of the cell/Maximum downlink cell bandwidth) | | | |

### 6.4.1.2 Using MML Commands

Before using MML commands, refer to **6.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

## Activation Command Examples

Activation command examples for UE-number-based idle mode MLB

```
//Turning on the MLB algorithm switch for a cell
MOD NRCELLALGOSWITCH: NrCellId=0, MlbAlgoSwitch=INTER_FREQ_IDLE_MLB_SW-1;
//Setting the MLB trigger mode and setting the idle MLB policy to the dedicated-priority-based policy
MOD NRCELLMLB: NrCellId=0, MlbTriggerMode=CELL_USER_NUM, IdleMlbPolicy=DEDICATED_PRIORITY;
//Setting the MLB evaluation period for the cell
MOD NRCELLMLB: NrCellId=0, MlbJudgePeriod=10;
//Setting the MlbDirection and MlbHoAdmissionThld parameters
MOD NRCELLMLB: NrCellId=0, MlbDirection=DOWNLINK, MlbHoAdmissionThld=95;
//Setting the InterFreqMlbUeNumThld, MlbUeNumOffset, and UeNumDiffThld parameters
MOD NRCELLMLB: NrCellId=0, InterFreqMlbUeNumThld=200, MlbUeNumOffset=50, UeNumDiffThld=25;
//Setting the InterFreqMlbUlUeNumThld, MlbUlUeNumOffset, and UlUeNumDiffThld parameters
MOD NRCELLMLB: NrCellId=0, InterFreqMlbUlUeNumThld=100, MlbUlUeNumOffset=20,
UlUeNumDiffThld=15;
//Enabling load information exchange with the local cell for a neighboring cell
MOD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=1, CellId=1, MlbHoFlag=TRUE,
NoHoFlag=PERMIT_HO;
//Enabling idle-mode MLB frequency optimization to avoid cell reselections to a neighboring frequency
when NRCellRelation.MlbHoFlag is set to FALSE for all cells on the neighboring frequency
MOD NRCELLMLB: NrCellId=0, MlbAlgoEnhSwitch=IDLE_MLB_FREQ_OPT_SW-1;
//Setting the ConnMlbUeNumThld parameter
MOD NRCELLMLB: NrCellId=0, ConnMlbUeNumThld=5;
//Configuring the T320 timer
MOD NRCELLRESELCONFIG: NrCellId=0, T320ForMlb=MIN10;
```

Activation command examples for spectral-efficiency-based idle mode MLB

```
//Turning on the MLB algorithm switch for a cell
MOD NRCELLALGOSWITCH: NrCellId=0, MlbAlgoSwitch=INTER_FREQ_IDLE_MLB_SW-1;
///Setting the MLB trigger mode, setting the idle MLB policy to the dedicated-priority-based policy, and
setting the cell load evaluation policy and cell capability scale factors
MOD NRCELLMLB: NrCellId=0, MlbTriggerMode=SPEC_EFF_BASED_USER_NUM, IdleMlbPolicy=
DEDICATED_PRIORITY, LoadEvaluationPolicy=DYNAMIC_EVAL, CellDlCapbScaleFactor=1,
CellUlCapbScaleFactor=1;
//Setting the MLB evaluation period for the cell
MOD NRCELLMLB: NrCellId=0, MlbJudgePeriod=10;
//Setting the MlbDirection and MlbHoAdmissionThld parameters
MOD NRCELLMLB: NrCellId=0, MlbDirection=DOWNLINK, MlbHoAdmissionThld=95;
//Setting the InterFreqMlbUeNumThld, MlbUeNumOffset, and UeNumDiffThld parameters
MOD NRCELLMLB: NrCellId=0, InterFreqMlbUeNumThld=200, MlbUeNumOffset=50, UeNumDiffThld=25;
//Setting the InterFreqMlbUlUeNumThld, MlbUlUeNumOffset, and UlUeNumDiffThld parameters
MOD NRCELLMLB: NrCellId=0, InterFreqMlbUlUeNumThld=100, MlbUlUeNumOffset=20,
UlUeNumDiffThld=15;
//Enabling load information exchange with the local cell for a neighboring cell
MOD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=1, CellId=1, MlbHoFlag=TRUE,
NoHoFlag=PERMIT_HO;
//Enabling idle-mode MLB frequency optimization to avoid cell reselections to a neighboring frequency
when NRCellRelation.MlbHoFlag is set to FALSE for all cells on the neighboring frequency
MOD NRCELLMLB: NrCellId=0, MlbAlgoEnhSwitch=IDLE_MLB_FREQ_OPT_SW-1;
//Setting the UE selection policy of MLB for the cell
MOD NRCELLMLB: NrCellId=0, MlbTriggerMode=SPEC_EFF_BASED_USER_NUM,
MlbUeSelectionPolicy=SA_USER-1&NSA_USER-0;
//Setting the ConnMlbUeNumThld parameter
MOD NRCELLMLB: NrCellId=0, ConnMlbUeNumThld=5;
```

```
//Configuring the T320 timer
MOD NRCELLRESELCONFIG: NrCellId=0, T320ForMlb=MIN10;
```

Activation command examples for radio-resource-based idle mode MLB

```
//Turning on the MLB algorithm switch for a cell
MOD NRCELLALGOSWITCH: NrCellId=0, MlbAlgoSwitch=INTER_FREQ_IDLE_MLB_SW-1;
///Setting the MLB trigger mode, setting the idle MLB policy to the dedicated-priority-based policy, and
setting the cell load evaluation policy and cell capability scale factors
MOD NRCELLMLB: NrCellId=0, MlbTriggerMode=RADIO_RESOURCE, IdleMlbPolicy= DEDICATED_PRIORITY,
LoadEvaluationPolicy=DYNAMIC_EVAL, CellDlCapbScaleFactor=1, CellUlCapbScaleFactor=1;
//Setting the MLB evaluation period for the cell
MOD NRCELLMLB: NrCellId=0, MlbJudgePeriod=10;
//Setting the MlbDirection and MlbHoAdmissionThld parameters
MOD NRCELLMLB: NrCellId=0, MlbDirection=DOWNLINK, MlbHoAdmissionThld=95;
//Setting the InterFreqMlbUeNumThld, MlbUeNumOffset, and UeNumDiffThld parameters
MOD NRCELLMLB: NrCellId=0, InterFreqMlbUeNumThld=200, MlbUeNumOffset=50, UeNumDiffThld=25;
//Setting the InterFreqMlbUlUeNumThld, MlbUlUeNumOffset, and UlUeNumDiffThld parameters
MOD NRCELLMLB: NrCellId=0, InterFreqMlbUlUeNumThld=100, MlbUlUeNumOffset=20,
UlUeNumDiffThld=15;
//Enabling load information exchange with the local cell for a neighboring cell
MOD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=1, CellId=1, MlbHoFlag=TRUE,
NoHoFlag=PERMIT_HO;
//Enabling idle-mode MLB frequency optimization to avoid cell reselections to a neighboring frequency
when NRCellRelation.MlbHoFlag is set to FALSE for all cells on the neighboring frequency
MOD NRCELLMLB: NrCellId=0, MlbAlgoEnhSwitch=IDLE_MLB_FREQ_OPT_SW-1;
//Setting the ConnMlbUeNumThld parameter
MOD NRCELLMLB: NrCellId=0, ConnMlbUeNumThld=5;
//Setting the MlbUlCceUsageThld and MlbDlCceUsageThld parameters
MOD NRCELLMLB: NrCellId=0, MlbTriggerMode=RADIO_RESOURCE, MlbUlCceUsageThld=30,
MlbDlCceUsageThld=30;
//Configuring the T320 timer
MOD NRCELLRESELCONFIG: NrCellId=0, T320ForMlb=MIN10;
```

### Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the MLB algorithm switch for the cell
MOD NRCELLALGOSWITCH: NrCellId=0, MlbAlgoSwitch=INTER_FREQ_IDLE_MLB_SW-0;
```

### 6.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

**NOTE**

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

## 6.4.2 Activation Verification

After low-frequency dedicated-priority-based idle mode MLB is enabled, dedicated priorities are delivered to UEs through the *CellReselectionPriorities* IE in RRCRelease messages.

On the MAE-Access, start a signaling trace management task with the parameters for **MLB Monitoring** under **Cell Performance Monitoring** specified to monitor MLB. The number of times idle mode MLB takes effect on the gNodeB can be monitored by **Idle Mlb Trigger Num**. If a non-zero value is returned, this function has taken effect.

### 6.4.3 Network Monitoring

Observe the value of the **N.User.RRCConn.Avg** counter to check whether the distribution of UEs in idle mode in the cells on different frequencies meets the expectation of MLB.

NOTE

As low-frequency dedicated-priority-based idle mode MLB does not take effect for UEs for which RRC connection releases are not triggered, the **N.User.RRCConn.Avg** counter value still differs among intra-sector cells, although the numbers of UEs among cells are balanced.

On the MAE-Access, start a signaling trace management task with the parameters for **MLB Monitoring** under **Cell Performance Monitoring** specified to monitor MLB. The number of times idle mode MLB takes effect on the gNodeB can be monitored by **Idle Mlb Trigger Num**.

# 7 High-Frequency UE-Number-based Idle Mode MLB (High-Frequency TDD)

## 7.1 Principles

High-frequency UE-number-based idle mode MLB balances the number of UEs in connected mode among common cells in a sector. When this function is enabled, for a UE about to be released and enter the idle mode, the gNodeB randomly selects a common cell that supports random access in a high-frequency sector as the target cell. The gNodeB then sends the frequency and frequency priority of the target cell to the UE through the *CellReselectionPriorities* IE in the RRCRelease message, and instructs the UE to reselect to the target cell.

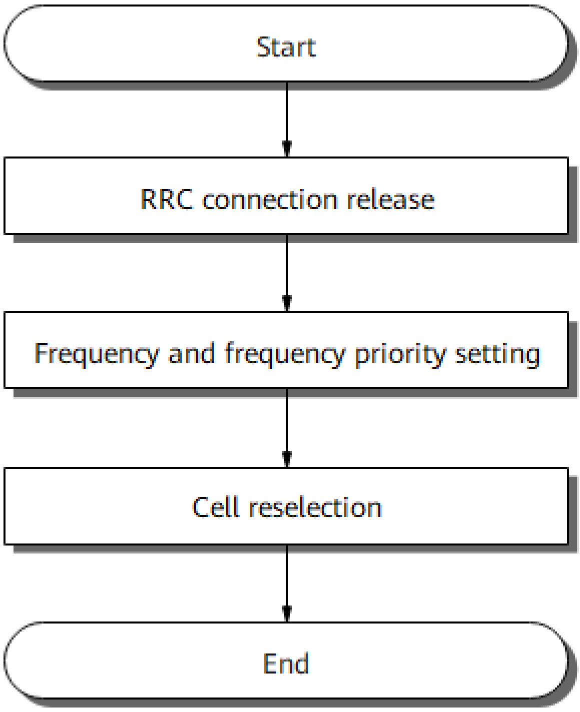
High-frequency UE-number-based idle mode MLB can take effect only if all of the following conditions are met:

- The **INTER_FREQ_IDLE_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter is selected for all common cells (cells with the **NRDUCell**.***SupplementaryCellInd*** parameter set to **NORMAL_CELL**) in the high-frequency sector.
- The cell-reselection priorities for the operating frequencies and neighboring frequencies of all common cells in the high-frequency sector are equal. The highest priority is recommended. To ensure that the cell-reselection priority of the operating frequency is the same as that of a neighboring frequency, the following sums of parameter values must be the same:
  - Sum of **NRCellReselConfig**.***CellReselPriority*** and **NRCellReselConfig**.***CellReselSubPriority***
  - Sum of **NRCellFreqRelation**.***CellReselPriority*** and **NRCellFreqRelation**.***CellReselSubPriority***

High-frequency UE-number-based idle mode MLB can take effect only for SA UEs in SA networking and NSA and SA hybrid networking.

**Figure 7-1** illustrates the procedure for high-frequency UE-number-based idle mode MLB.

**Figure 7-1** MLB procedure




## RRC Connection Release

The gNodeB delivers the frequency and frequency priority of the target cell through the *CellReselectionPriorities* IE in the RRCRelease message. The *CellReselectionPriorities* IE includes the following information:

- Frequencies and their dedicated priorities
- Timer T320, which specifies the validity time for the dedicated cell-reselection priorities. The timer length is fixed to 120 minutes.

After the RRC connection of a UE is released, the UE enters the idle mode.

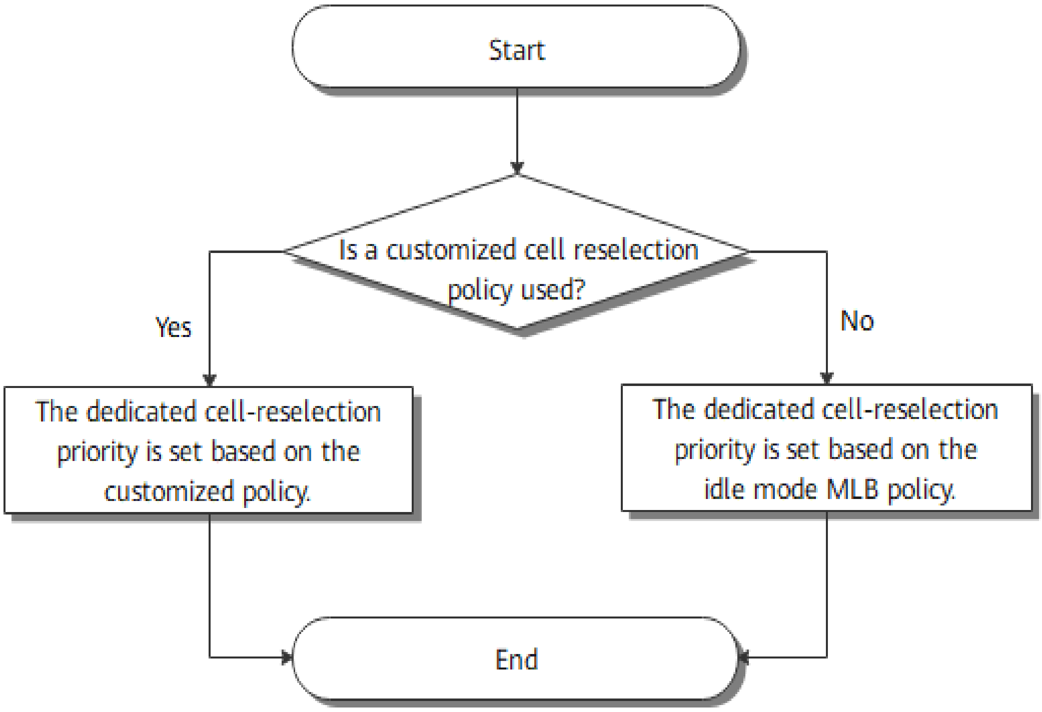
NOTE

High-frequency UE-number-based idle mode MLB does not take effect for UEs for which the RRC connection releases are not triggered.

## Setting Frequencies and Frequency Priorities

**Figure 7-2** illustrates the procedure for setting frequencies and frequency priorities

**Figure 7-2** Procedure for setting frequencies and frequency priorities




The gNodeB checks whether the UE uses a customized cell reselection policy, that is, whether the UE is configured with an RFSP-specific dedicated cell-reselection priority.

- If the UE uses the customized cell reselection policy, the gNodeB delivers the RFSP-specific dedicated cell-reselection priority.
- If the UE does not use the customized cell reselection policy, the gNodeB sets the cell-reselection frequency priority based on the UE-number-based idle mode MLB policy. In this policy, the gNodeB randomly selects an intra-sector high-frequency cell and delivers the operating frequencies and frequency priorities of this cell, inter-sector (both high-frequency and low-frequency) inter-frequency cells, and inter-RAT cells to the UE.

### Cell Reselection

The UE reselects to the target cell based on the target frequencies and their priorities delivered from the gNodeB. For details about cell reselection, see, *SA Mobility Management in Idle Mode*.

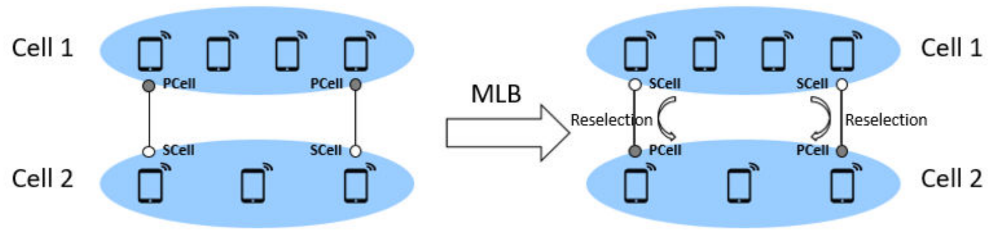
# 7.2 Network Analysis

## 7.2.1 Benefits

High-frequency UE-number-based idle mode MLB balances the number of UEs in connected mode among intra-sector common cells, thereby balancing the resource usage among these cells. The greater the difference in the number of UEs among intra-sector cells, the more the gains.

In CA scenarios, if the PCells of large-packet UEs change after MLB is performed, but the cell set to which large-packet UEs are reselected do not change, this function provides no gains, as shown in **Figure 7-3**. Therefore, the more the large-packet UEs, the smaller the gains.

**Figure 7-3** Diagram of MLB for CA UEs




When the service types differ among intra-sector common cells, and cells with a large number of large-packet services are overloaded, high-frequency UE-number-based idle mode MLB is not recommended in this scenario.

## 7.2.2 Impacts

The impacts between this function and other functions are described in **7.3.2.3 Function Impacts**.

After high-frequency UE-number-based idle mode MLB is enabled, the values of the following counters for intra-sector FR2 cells will become relatively balanced.

**Table 7-1** MLB-related counters for FR2 cells

| Counter ID | Counter Name |
| --- | --- |
| 1911816771 | **N.User.RRCConn.Avg** |
| 1911816773 | **N.User.CA.PCell.DL.Avg** |
| 1911816775 | **N.User.CA.PCell.UL.Avg** |

NOTE

As high-frequency UE-number-based idle mode MLB does not take effect for UEs for which the RRC connection releases are not triggered, this function can balance the numbers of UEs among cells. However, the **N.User.RRCConn.Avg** counter value differs among intra-sector cells.

The larger the value of the **NRDUCellQciBearer**.***UeInactivityTimer*** parameter, the lower the probability that UEs enter idle mode, and the lower the gains of high-frequency UE-number-based idle mode MLB. For details about the UE inactivity timer, see *5G Networking and Signaling*.

# 7.3 Requirements

## 7.3.1 Licenses

None

## 7.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

### 7.3.2.1 Prerequisite Functions

None

### 7.3.2.2 Mutually Exclusive Functions

None

### 7.3.2.3 Function Impacts

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| High-frequency TDD | RFSP-based camping and handover policies | **gNBRfspConfig**.***gNBFreqPriorityGroupId*** | *Flexible User Steering* | High-frequency UE-number-based idle mode MLB (controlled by the **INTER_FREQ_IDLE_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter) does not take effect for UEs configured with RFSP-specific dedicated cell-reselection priorities. |

## 7.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

### Base Station Models

5900 series base stations (macro base stations) and DBS5900 LampSite base stations

### Boards

All NR-capable main control boards and NR TDD mmWave baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

### RF Modules

For macro base stations, all of the NR TDD-capable AAUs that work in high frequency bands support this function. For details, see the technical specifications of AAUs in *3900 & 5900 Series Base Station Product Documentation*.

For LampSite base stations, the pRRU5551JR supports this function. For details, see *LampSite pRRU Technical Specifications* in *3900 & 5900 Series Base Station Product Documentation*.

## 7.3.4 Networking

There must be at least two frequencies available on the live network.

## 7.3.5 Others

None

# 7.4 Operation and Maintenance

## 7.4.1 Data Configuration

### 7.4.1.1 Data Preparation

Before activating this function, ensure that the **CellReselPriority** and **NRCellFreqRelation** MOs have been configured. For details about the configuration, see *SA Mobility Management in Idle Mode* and *NSA Mobility Management*.

**Table 7-2** describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

**Table 7-2** Parameters used for activation

| Parameter Name | Parameter ID | Option | Setting Notes |
|---|---|---|---|
| MLB Algorithm Switch | **NRCellAlgoSwitch**.***MlbAlgoSwitch*** | INTER_FREQ_IDLE_MLB_SW | Select this option for all common cells (cells with the **NRDUCell**.***SupplementaryCellInd*** parameter set to **NORMAL_CELL**) in a sector. |
| Cell Reselection Priority | **NRCellReselConfig**.***CellReselPriority*** | None | The value of this parameter must be determined together with the priorities of other frequencies during network planning. |
| Cell Reselection Sub-Priority | **NRCellReselConfig**.***CellReselSubPriority*** | None | The value of this parameter must be determined together with the priorities of other frequencies during network planning. |
| Cell Reselection Priority | **NRCellFreqRelation**.***CellReselPriority*** | None | The value of this parameter must be determined together with the priorities of other frequencies during network planning. |
| Cell Reselection Sub-Priority | **NRCellFreqRelation**.***CellReselSubPriority*** | None | The value of this parameter must be determined together with the priorities of other frequencies during network planning. |
| Note: The cell-reselection priorities for the operating frequencies of all common cells (cells with the **NRDUCell**.***SupplementaryCellInd*** parameter set to **NORMAL_CELL**) in the high-frequency sector are equal. The highest priority is recommended. Equal cell-reselection priorities mean that the sum of the **NRCellReselConfig**.***CellReselPriority*** and **NRCellReselConfig**.***CellReselSubPriority*** parameter values is equal to the sum of the **NRCellFreqRelation**.***CellReselPriority*** and **NRCellFreqRelation**.***CellReselSubPriority*** parameter values. | | | |

### 7.4.1.2 Using MML Commands

Before using MML commands, refer to **7.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

## Activation Command Examples

```
//Turning on the MLB algorithm switch for all common cells in a high-frequency sector and setting the cell-reselection priorities
MOD NRCELLALGOSWITCH: NrCellId=7, MlbAlgoSwitch=INTER_FREQ_IDLE_MLB_SW-1;
MOD NRCELLALGOSWITCH: NrCellId=8, MlbAlgoSwitch=INTER_FREQ_IDLE_MLB_SW-1;
MOD NRCELLRESELCONFIG: NrCellId=7, CellReselPriority=7, CellReselSubPriority=0DOT0;
MOD NRCELLFREQRELATION: NrCellId=7, SsbFreqPos=22476, FrequencyBand=N257, CellReselPriority=7, CellReselSubPriority=0DOT0;
MOD NRCELLRESELCONFIG: NrCellId=8, CellReselPriority=7, CellReselSubPriority=0DOT0;
MOD NRCELLFREQRELATION: NrCellId=8, SsbFreqPos=22487, FrequencyBand=N257, CellReselPriority=7, CellReselSubPriority=0DOT0;
```

## Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the MLB algorithm switch for all common cells in a high-frequency sector
MOD NRCELLALGOSWITCH: NrCellId=7, MlbAlgoSwitch=INTER_FREQ_IDLE_MLB_SW-0;
MOD NRCELLALGOSWITCH: NrCellId=8, MlbAlgoSwitch=INTER_FREQ_IDLE_MLB_SW-0;
```

### 7.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

**NOTE**

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

## 7.4.2 Activation Verification

After high-frequency UE-number-based idle mode MLB is enabled, if it is observed that the CellReselectionPriorities IE in the RRCRelease message contains the information about the frequencies for cell reselection and their priorities and the values of the following counters become relatively balanced among intra-sector FR2 cells, this function has taken effect.

**Table 7-3** MLB-related counters for FR2 cells

| Counter ID | Counter Name |
|---|---|
| 1911816771 | **N.User.RRCConn.Avg** |
| 1911816773 | **N.User.CA.PCell.DL.Avg** |
| 1911816775 | **N.User.CA.PCell.UL.Avg** |

## 7.4.3 Network Monitoring

Observe the value of the **N.User.RRCConn.Avg** counter to check whether the distribution of UEs in connected mode in the cells on different frequencies meets the expectation of MLB.

**NOTE**

As high-frequency UE-number-based idle mode MLB does not take effect for UEs for which the RRC connection releases are not triggered, this function can balance the numbers of UEs among cells. However, the **N.User.RRCConn.Avg** counter value differs among intra-sector cells.

# 8 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

**NOTE**

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

**FAQ 1: How do I find the parameters related to a certain feature from parameter reference?**

**Step 1** Open the EXCEL file of parameter reference.

**Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

**Step 3** Click **OK**. All parameters related to the feature are displayed.

**----End**

**FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?**

**Step 1** Open the EXCEL file of the used reserved parameter list.

**Step 2** On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

**----End**

# 9 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

**NOTE**

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

**FAQ: How do I find the counters related to a certain feature from performance counter reference?**

**Step 1** Open the EXCEL file of performance counter reference.

**Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

**Step 3** Click **OK**. All counters related to the feature are displayed.

**----End**

# 10 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

# 11 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- 3GPP TS 38.423: "NG-RAN; Xn Application Protocol (XnAP)"
- 3GPP TS 38.314: "NR; Layer 2 measurements"
- *Flexible User Steering*
- *Mobility Load Balancing in Connected Mode*
- *SA Mobility Management in Connected Mode*
- *SA Mobility Management in Idle Mode*
- *NSA Mobility Management*
- *FWA*
- *License Control Item Lists*
- *5G Networking and Signaling*
- *High Speed Mobility*
- *CA Experience Improvement*
- *Technical Specifications* in *3900 & 5900 Series Base Station Product Documentation*